

lightning arrester type 1, requirement class B, Uc 350 V, protective modules, plug-in, 4-pole, 3+1 circuit for TN-S and TT systems, with remote display



| General data                                                          |                                       |
|-----------------------------------------------------------------------|---------------------------------------|
| standard                                                              | IEC 61643-11: 2011, EN 61643-11: 2012 |
| product designation                                                   | Surge protection device               |
| <b>SPD classification according to EN 61643-11</b>                    |                                       |
| • Test Class I, Type 1                                                | Yes                                   |
| • Test Class II, Type 2                                               | Yes                                   |
| • Test Class III, Type 3                                              | No                                    |
| number of SPD ports                                                   | 1                                     |
| design of the product                                                 | Lightning arresters                   |
| design of pole                                                        | 3+N/PE                                |
| designation of the protective paths                                   | L-N, L-PE, N-PE                       |
| accessories                                                           | 3 x 5SD7418-1 + 1 x 5SD7418-0         |
| fastening method                                                      | TH35 DIN rail                         |
| material of the enclosure                                             | PBT                                   |
| size of surge arrester                                                | 8 TE                                  |
| degree of pollution                                                   | 2                                     |
| overvoltage category according to IEC 61010-1                         | III                                   |
| protection class IP at connection all terminals                       | IP20                                  |
| shock acceleration                                                    | 25 gn                                 |
| vibrational acceleration at 5 Hz ... 500 Hz limited to 2,5 h per axis | 5 gn                                  |
| relative humidity during operation                                    | 5 ... 95 %                            |
| installation altitude at height above sea level maximum               | 2 000 m                               |
| width                                                                 | 142.4 mm                              |
| height                                                                | 95 mm                                 |
| depth                                                                 | 71.5 mm                               |
| net weight                                                            | 1 560 g                               |
| Electrical data                                                       |                                       |
| type of distribution system                                           | TT, TN-S                              |
| <b>operating voltage</b>                                              |                                       |
| • at AC                                                               | 230 V                                 |
| value range of the operating frequency                                | 50 / 60 Hz                            |
| <b>continuous operating voltage</b>                                   |                                       |
| • at AC maximum                                                       | 350 V                                 |
| • between N and PE at AC maximum                                      | 350 V                                 |
| • between L and (PE)N at AC maximum                                   | 350 V                                 |
| <b>discharge current</b>                                              |                                       |
| • between L and (PE)N at (8/20) µs                                    | 25 kA                                 |
| • between L and N at (8/20) µs                                        | 50 kA                                 |
| • between L and PE at (8/20) µs                                       | 25 kA                                 |

|                                                                                       |                                   |
|---------------------------------------------------------------------------------------|-----------------------------------|
| • between N and PE at (8/20) $\mu$ s                                                  | 50 kA                             |
| • between N and PE at (8/20) $\mu$ s                                                  | 100 kA                            |
| total lightning impulse current at (10/350) $\mu$ s                                   | 100 kA                            |
| <b>lightning current peak value at (10/350) <math>\mu</math>s</b>                     |                                   |
| • lightning current peak value between L and PE                                       | 25 kA                             |
| • lightning current peak value between N and PE                                       | 100 kA                            |
| • lightning current peak value between L and N                                        | 25 kA                             |
| <b>charge of the flash at (10/350) <math>\mu</math>s</b>                              |                                   |
| • charge of the flash between L and N                                                 | 12.5 A·s                          |
| • charge of the flash between L and PE                                                | 12.5 A·s                          |
| • charge of the flash between N and PE                                                | 50 A·s                            |
| <b>specific energy of the flash at (10/350) <math>\mu</math>s</b>                     |                                   |
| • between L and N                                                                     | 160 kJ/?                          |
| • between L and PE                                                                    | 160 kJ/?                          |
| • between N and PE                                                                    | 2 500 kJ/?                        |
| <b>follow current extinguishing capability</b>                                        |                                   |
| • between N and PE                                                                    | 100 A                             |
| • between L and N                                                                     | 50 kA                             |
| short-circuit rating (SCCR) at 264 V                                                  | 50 kA                             |
| <b>protection level</b>                                                               |                                   |
| • between L and N maximum                                                             | 1.5 kV                            |
| • between L and PE maximum                                                            | 2.5 kV                            |
| • between N and L                                                                     | 1.5 kV                            |
| • between N and PE maximum                                                            | 1.5 kV                            |
| • between PE and N and/or L                                                           | 1.5 kV                            |
| <b>residual voltage</b>                                                               |                                   |
| • between L and (PE)N<br>— at rated value of discharge current maximum                | 1.5 kV                            |
| • between L and PE<br>— at rated value of discharge current maximum                   | 2.5 kV                            |
| • between N and PE<br>— at rated value of discharge current maximum                   | 1.5 kV                            |
| <b>response value of the surge voltage at 6 kV at (1.2/50) <math>\mu</math>s</b>      |                                   |
| • between L and N                                                                     | 1.5 kV                            |
| • between L and PE                                                                    | 2.5 kV                            |
| • between N and PE                                                                    | 1.5 kV                            |
| • response time between L and (PE)N                                                   | 100 ns                            |
| • response time between N and PE                                                      | 100 ns                            |
| adjustable response factor of tripping current                                        | 1.6                               |
| fuse protection type at V-shaped connection                                           | 125 A AC (gG)                     |
| fuse protection type for T-connector                                                  | 315 A AC (gG)                     |
| <b>Connections/ Terminals</b>                                                         |                                   |
| type of electrical connection                                                         | Screw terminal                    |
| stripped length                                                                       | 18 mm                             |
| tightening torque                                                                     | 4.3 ... 4.7 N·m                   |
| <b>connectable conductor cross-section</b>                                            |                                   |
| • for finely stranded conductor                                                       | 2.5 ... 25 mm <sup>2</sup>        |
| • for rigid conductor                                                                 | 2.5 ... 35 mm <sup>2</sup>        |
| • finely stranded                                                                     | 2.5 ... 25 mm <sup>2</sup>        |
| AWG number as coded connectable conductor cross section                               | 13 ... 2                          |
| design of the thread of the connection screw                                          | M5                                |
| signal design                                                                         | Optical, remote signaling contact |
| <b>Indicator/remote signaling</b>                                                     |                                   |
| product component remote signaling contact                                            | Yes                               |
| switching function of the remote signaling contacts                                   | PDT contact                       |
| operating voltage of the remote signaling contacts at AC                              | 12 ... 250 V                      |
| operational current of the remote signaling contacts at AC                            | 10 mA ... 1 A                     |
| connection type of remote signaling contact                                           | M2                                |
| connectable conductor cross-section for remote signaling contacts for rigid conductor | 0.14 ... 1.5 mm <sup>2</sup>      |

|                                                                                                                                                                                                                                                                                    |                                                                                                           |                          |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------|-------|
| connectable conductor cross-section for remote signaling contacts for finely stranded conductor                                                                                                                                                                                    | 0.14 ... 1.5 mm <sup>2</sup>                                                                              |                          |       |
| AWG number as coded connectable conductor cross section for remote signaling contacts                                                                                                                                                                                              | 28 ... 15                                                                                                 |                          |       |
| tightening torque for remote signaling contacts                                                                                                                                                                                                                                    | 0.25 N·m                                                                                                  |                          |       |
| stripped length of the cable for remote signaling contacts                                                                                                                                                                                                                         | 7 mm                                                                                                      |                          |       |
| <b>NEMA/UL - Data</b>                                                                                                                                                                                                                                                              |                                                                                                           |                          |       |
| type of surge protective device (SPD) according to UL                                                                                                                                                                                                                              | 4CA                                                                                                       |                          |       |
| type of distribution system according to UL                                                                                                                                                                                                                                        | 3Y                                                                                                        |                          |       |
| type of distribution system                                                                                                                                                                                                                                                        | TT, TN-S                                                                                                  |                          |       |
| designation of the protective paths according to UL                                                                                                                                                                                                                                | L-L, L-N, L-G, N-G                                                                                        |                          |       |
| <b>TOV behavior</b>                                                                                                                                                                                                                                                                |                                                                                                           |                          |       |
| <ul style="list-style-type: none"> <li>at TOV test voltage (L-N)</li> <li>at TOV test voltage (N-PE)</li> </ul>                                                                                                                                                                    | 415 V AC (5 s / withstand mode) / 457 V AC (120 min / withstand mode)<br>1200 V (200 ms / withstand mode) |                          |       |
| <b>Measured Limiting Voltage (MLV)</b>                                                                                                                                                                                                                                             |                                                                                                           |                          |       |
| <ul style="list-style-type: none"> <li>between L and L</li> <li>between L and Ground (GND)</li> <li>between L and N</li> <li>between N and Ground (GND)</li> </ul>                                                                                                                 | 2.45 kV<br>1.57 kV<br>1.35 kV<br>1.08 kV                                                                  |                          |       |
| <b>Maximum Continuous Operating Voltage (MCOV)</b>                                                                                                                                                                                                                                 |                                                                                                           |                          |       |
| <ul style="list-style-type: none"> <li>between L and L</li> <li>between L and Ground (GND)</li> <li>between L and N</li> <li>between N and Ground (GND)</li> </ul>                                                                                                                 | 528 V<br>528 V<br>264 V<br>264 V                                                                          |                          |       |
| <b>discharge current</b>                                                                                                                                                                                                                                                           |                                                                                                           |                          |       |
| <ul style="list-style-type: none"> <li>between N and Ground (GND) according to UL rated value</li> <li>between L and N according to UL rated value</li> <li>between L and Ground (GND) according to UL rated value</li> <li>between L and L according to UL rated value</li> </ul> | 20 kA<br>20 kA<br>20 kA<br>20 kA                                                                          |                          |       |
| <b>AWG number as coded connectable conductor cross section</b>                                                                                                                                                                                                                     |                                                                                                           |                          |       |
| <ul style="list-style-type: none"> <li>according to UL</li> <li>for remote signaling contacts according to UL</li> </ul>                                                                                                                                                           | 12 ... 2<br>30 ... 14                                                                                     |                          |       |
| operating voltage of the remote signaling contacts according to UL                                                                                                                                                                                                                 | 125 V                                                                                                     |                          |       |
| operational current of the remote signaling contacts at AC according to UL                                                                                                                                                                                                         | 1 A                                                                                                       |                          |       |
| <b>ambient temperature</b>                                                                                                                                                                                                                                                         |                                                                                                           |                          |       |
| <ul style="list-style-type: none"> <li>during operation</li> <li>during storage</li> </ul>                                                                                                                                                                                         | -40 ... +80 °C<br>-40 ... +80 °C                                                                          |                          |       |
| installation altitude above sea level according to UL                                                                                                                                                                                                                              | 6 562 ft                                                                                                  |                          |       |
| gross weight [lb] according to UL                                                                                                                                                                                                                                                  | 3.56 lb(av)                                                                                               |                          |       |
| net weight [lb] according to UL                                                                                                                                                                                                                                                    | 3.16 lb(av)                                                                                               |                          |       |
| combustibility class according to UL 94                                                                                                                                                                                                                                            | V0                                                                                                        |                          |       |
| standards according to UL                                                                                                                                                                                                                                                          | UL 1449 edition 4                                                                                         |                          |       |
| <b>Approvals Certificates</b>                                                                                                                                                                                                                                                      |                                                                                                           |                          |       |
| <table> <tr> <td>General Product Approval</td><td>other</td></tr> </table>                                                                                                                                                                                                         |                                                                                                           | General Product Approval | other |
| General Product Approval                                                                                                                                                                                                                                                           | other                                                                                                     |                          |       |



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|-------|-------------|
| other | Environment |
|-------|-------------|

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## Further information

### Information on the packaging

<https://support.industry.siemens.com/cs/ww/en/view/109813875>

### Information- and Downloadcenter (Catalogs, Brochures,...)

<http://www.siemens.com/lowvoltage/catalogs>

### Industry Mall (Online ordering system)

<https://mall.industry.siemens.com/mall/en/en/Catalog/product?mlfb=5SD7414-1>

### Service&Support (Manuals, Certificates, Characteristics, FAQs,...)

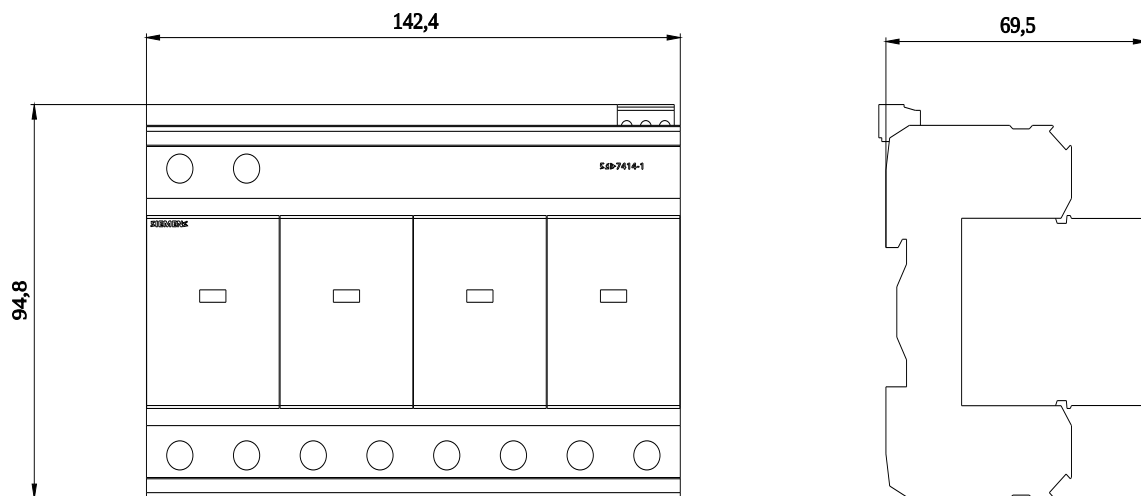
<https://support.industry.siemens.com/cs/ww/en/ps/5SD7414-1>

### Image database (product images, 2D dimension drawings, 3D models, device circuit diagrams, ...)

[http://www.automation.siemens.com/bilddb/cax\\_en.aspx?mlfb=5SD7414-1](http://www.automation.siemens.com/bilddb/cax_en.aspx?mlfb=5SD7414-1)

### CAX-Online-Generator

<http://www.siemens.com/cax>



last modified:

8/7/2025 

